

Institute of Actuaries of India

Subject ST1 – Health and Care Insurance Specialist Technical

May 2014 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**a) No worse terms**

In certain circumstances, on renewal of a PMI policy, a new insurer will agree to offer cover at least as comprehensive and with no additional underwriting conditions as the policyholder's current policy; the renewal or "switch" is said to be accepted on no worse terms.

b) Capitation

This term relates to the practice of charging for cover by forecasting the likely claims on an individual basis adjusted for expenses and profit, as the premium. It is used by insurance companies to provide a set of medical benefits (like dental claims or mental health claims) and pass the risk to the provider by giving a proportion of premium as an up-front for each managed person.

c) Community rating

Community rating most often refers to the practice of charging all policyholders (or a significant subset of the persons insured) the same premium rate irrespective of rating factors applicable to individual such as age, gender and medical history.

d) Personal capability assessment

This is an alternative way of assessing disability that is largely independent of the age and the occupation (if any) of the person being assessed. Personal capability assessment involves assessment of the ability of an individual to complete everyday tasks of living (e.g. climbing stairs, bending, lifting and carrying). This is used as a test to access level of incapacity under Income Protection Policy.

[4 Marks]

Solution 2:**i) Benefits covered under a standard private medical insurance include:**

- In-patient benefits requiring stay overnight in hospitals
- Specified Day care procedures
- Pre hospitalization expenses
- Post hospitalization expenses
- Domiciliary expenses
- Ambulance expenses
- Health check-ups (in few cases)

Exclusions:

- Pre-existing diseases
- Cosmetic surgeries
- Mental disorders
- Maternity related claims
- Acts of adventure/sports leading to injuries or accidents
- AIDS/HIV
- Dental cover
- Optical cover

[3]

- ii) Private medical insurance combined with critical illness could be a good proposition with given advantages and disadvantages

For customers:

Advantages:

- Wider coverage with chronic ailment cases for listed diseases also covered.
- Lower premium rates than if both products bought separately as the combo product would have lower expense margins built in to the premium.
- No need to purchase multiple policies for different healthcare needs.

Disadvantages:

- Not all chronic ailments would be covered which limits the benefits provided under the combo product
- Pre-existing chronic ailments may not be covered
- May not understand the cover being provided/TCF issues

For insurance company:

Advantages:

- Higher product volume and revenue generation
- Competitive edge if peers are not selling the similar kind of product
- Reduced administrative expenses when the two products are sold as combo- products
- Distributors may like the new products

Disadvantages:

- system inefficiency of managing combo policies
- regulatory delays and product filing hassles
- higher dispute at the time of claims (reputation risks)
- covered critical illnesses may not be rated highly (or thought to be worth the extra cost of cover) resulting in lower volumes
- possible effects on availability/cost of reinsurance
- may need to train staff, distributors etc

[4]

[Total Marks-7]

Solution 3:**Advantages:**

- Larger agency workforce to leverage on
- Increases competition among agents resulting in higher efficiency and better customer service
- Increase reach to the market and able to bring in low risk and more profitable customers
- Higher expected product sales and volumes
- Competitive edge in selling health insurance products with innovative features through agency distribution channel
- Wider choice of products for agents to provide solutions to their customers and improve agents' productivity and income. This would also help reducing agents attrition.

Disadvantages:

- Training and management of new agents may create administrative and operational constraints
- Higher expenses associated with training and licensing/registration of agents
- Managing agents expectations (may want high commission)
- Customer service issues if agents do not understand the complexity of health products fully.
- Increase in fraudulent claims where agents themselves get involved to meet their financial interests.
- High attrition rates for agents and mis-selling issues would pose challenge.

[Total Marks-6]

Solution 4:

- i. No claim discount of 10% of premium for each claim free year which can accumulate maximum up to 50%. On claim, the discount decreases in the same manner as it was increased.

Positives:

- It will encourage the insured to pay for small claims, thus reducing the number of claims and keeping down claims administration costs.
- It may encourage the insured to renew his/her policy, particularly if the discount is not transferable between insurers. This would reduce risk of anti-selection.
- the no claim discount is not reduced to zero after a claim, this may help to increase sales/renew policies and volume if such benefit is not offered by other cos. Higher volume would result in to lower cost.

Negatives:

- It may be viewed as immoral by the industry because it discourages people from claiming when they need treatment.
- It will increase other administration costs. We need to ensure that our systems can deal with the ratings imposed.
- May be administratively difficult to handle

[2]

ii. Excess based on the level of sum insured opted by insuredPositives:

- This would act to reduce the number of claims, so reducing the cost of claims administration.
- Overall claims costs will reduce, allowing premiums to be more competitive compared to those with “flat” excess levels

Negatives:

- It may look bad if the excess is seen to discourage people from seeking the treatment they need.
- Customers may not understand how the excess operates

[2]

iii. Limit on maximum claims payments in a yearPositives:

- It will limit the insurer’s liability, resulting in lower claims overall.
- Premiums can be reduced to more competitive levels, and this should lead to more sales and lower per policy cost

Negatives:

- If the insured is seriously ill (eg needs an organ transplant), the insurance cover may run out before the treatment is complete. There is a risk of bad publicity harming the insurer’s reputation.
- This would have most effect at older ages, where claims are more frequent. The insured may feel that he or she has paid for many years without claiming, and so any restriction may appear to be unfair (in spite of the policy wording).
- The insurer will still have to pay all claims, so the claims administration expenses will be high.
- The overall impact is likely to be small because most cases where expensive or lengthy treatment is required will be for chronic conditions, which are excluded from most PMI policies.

[2]

[Total Marks-6]

Solution 5:

Possible government actions:

- Restrict the sale of product which are giving higher losses
- Insurer may require to include certain waiting period and exclusion in the terms and conditions
- May restrict the sale of certain products to certain sales channel only or agent whose claim experience is better.
- May bring regulation to standardize the practices of health service providers to bring more certainty about cost of claim.
- May bring regulation for third party administration to standardize to claim management process and cost.
- May prescribe minimum pricing standard to ensure adequacy of premium rates.
- May require differential pricing for different segments to exactly reflect the claim experience of that segment.
- May prescribe standard definition for illnesses and claims to avoid misinterpretation risk.
- Prescribe higher training requirements for the agents selling the health products.
- May prescribe minimum underwriting standards for the underwriters to select lives

- Companies are asked to have a mandatory fraud identification framework in place to contain such cases and bring down loss ratios.

- May prescribe cap on fees for the health service provider to restrict the cost
- Regulator may restrict the guarantee on premium rates or may require more frequent review of premium rates
- May penalize the companies whose loss ratio is higher
- May prescribe minimum risk to be reinsured for each policy in order to get benefit of diversification and solvency relief.
- May reduce solvency requirement to keep premium rates lower to support growth.
- May provide cross subsidy in premium rates where premium rates are kept low to provide health cover for social sector.
- To encourage people to buy health insurance, government may increase tax benefits of health premiums. This would help to encourage people to buy health cover and also reduce the anti selection risks.
- May require compulsory claim cost sharing with the claimant to reduce frauds or mandatory no-claim bonus for policyholders.
- May cap first year commission and require higher renewal commission to improve persistency and ensure that good lives also stay longer.

[10 Marks]

Solution 6:**i.****Advantage:**

The company will no longer incur the costs of obtaining the additional medical information. This savings will give the opportunity for premiums to be reduced, or alternatively profits increased.

The company would normally wish to reduce premiums if it can. This will make the contract more competitive, and thus should increase sales.

The underwriting process should be faster. This will be popular with prospective policyholders and with insurance intermediaries. It should lead to increased sales.

Disadvantage:

Reinsurers may wish to re-price reinsurance cover, or become more involved in the underwriting process for large cases. This might increase costs or introduce delays, offsetting the benefits above.

The underwriters will have to base decisions on more limited information and so more experienced underwriting staff may be needed, increasing staff costs and reducing the expense savings.

The company could increase the number and depth of questions on the application form. This may be a deterrent to prospective policyholders, and offset the benefit of the faster underwriting process.

With less information, it may be necessary to decline more lives. This may lead to complaints, or intermediaries not recommending the company.

The risk of underwriting incorrectly will be increased. This may cause claim experience to worsen. Alternatively unnecessarily harsh terms might be imposed, to ensure experience is no worse. This may have the same consequences as increased declinations.

There is a risk of anti-selection if the company attracts lives where the additional information from a medical report would cause other companies to impose special terms.

There is a risk of moral hazard if customers believe they can give incorrect information and not be found out by a medical exam.

[7]**ii.** The company would wish to investigate experience against assumptions in:

- Critical illness rates
- expenses
- new business volumes
- market share
- the proportion of cases where special terms were offered
- the rate of acceptance of special terms
- the proportion of cases declined
- intermediaries perception of the advantages/disadvantages of the new arrangements
- the number of, and volume of business from, supporting intermediaries.

With only two years data, any changes in claim experience are very unlikely to be statistically significant unless the company is very large, and most mortality tables have a select period of at least two years.

Nevertheless the claim rate within the first and second years of the policy could be calculated both before and after the change.

The number of reported critical illness and the exposed to risk would be counted, taking care that they match. If there is sufficient data to generate credible results, analysis by cause of illness would be a useful guide to the underwriters.

Analysis by age and sex might be possible, but for critical illness reported at short duration this would be less useful than the cause of critical illness analysis.

Underwriting expenses should emerge from the company's expense analysis.

Before the change these would include costs paid for medical reports and examinations.

The costs of the underwriting department would be determined. The direct costs would be grossed up for management costs and other overheads. The costs could be expressed both as a charge per new policy written and as a charge per case proposed.

For comparison with the costs two years ago, it is important to inflation adjust the results.

It will be important to look at new business both in terms of the number of cases and the average size.

Changes in the way industry statistics are collected for market share results will have to be allowed for.

The company could also investigate whether any competitors have adopted similar underwriting approaches.

Most of the other statistics are a simple analysis of the data from the company's systems. Consistency of approach is important.

A questionnaire could be prepared for intermediaries that have supported the company both before and after the change, and for those intermediaries who either started or stopped supporting the company as a result of the change.

[10]

[Total Marks-17]

Solution 7:

- i. Below mentioned features could be considered:
 - Cashless coverage would be important for poor people so as to avoid out of pocket expenses. This should therefore be considered.
 - Exclusions and limits should be minimal since target population would not understand them. However, the insurance company must control its risk exposure.
 - Maternity benefits may be covered since childbirth often leads to sickness among the poor.
 - Income replacement may be covered since the insured population is unlikely to have savings off which to survive during periods of illness.

- Health care providers must be easily accessible, even in remote areas covered by the scheme.
- Outpatient care should be provided. It can act as an early warning and a gatekeeper for in-patient care, thus helping reduce costs overall.

[3]

- ii.** The first and the foremost requirement would be to access the eligible population for the desired benefits. This may be facilitated by working with a local agency, e.g. a local NGO with good contacts and reach in each area.

How are families assessed as to whether they are BPL.

How often is this measured (annually/monthly etc) or perhaps retrospectively (e.g. income over last year)

What happens if circumstances change

Definition of a family

It is important that a high proportion of those targeted enroll so as to reduce the risk of anti-selection.

NGO's existing infrastructure could be used for administration which would be a cost-efficient option.

Enrolment on a family basis rather than an individual basis should be promoted to reduce anti-selection.

The window for joining the scheme should be limited rather than continuous.

The scheme should be widely publicized to ensure large participation.

[3]

- iii.** Following challenges would be faced by the interested insurance cos.:

- To maintain financial strength on realistic and regulatory bases
- Lack of experience of group health business
- Lack of experience of rural business
- Requirement for long-term commitment to the scheme so that there will be an expectation of continuing cover. This may require a business plan that demonstrates the company's financial incentive, i.e. long-term profitability, to develop the business.
- Initial set up costs in reaching the families
- Ongoing costs would need to be kept to a minimum
- May be difficult to enroll geographically remote families
- Need to ensure large participation to reduce risk of anti-selection
- Restriction on future premium revision (if required)
- Political interference may limit underwriting capabilities

- Potential need to promote brand awareness
- Are there competitors or would the insurer have a monopoly if chosen by the government to provide this service.

[4]

iv. The company may:

- 1) Use data from any other relevant source, e.g. NGO, national statistics
- 2) Any suitable overseas data if similar products set up in other countries
- 3) Collect all data and analyze its own experience frequently
- 4) Would need to allow for trends and expenses
- 5) Arrange reinsurance
- 6) Ensure that adequate capital is in place to deal with claims volatility or mis-estimation risk
- 7) Maintain open dialogue with the state government and the regulator

[2]

[Total Marks-12]

Solution 8:

i) Results from experience studies can be used in following ways:

- improving the pricing basis/ re-pricing the products
- establishing / revising the reserving basis
- change the investment strategy
- change the reinsurance strategy
- changing / improving the marketing message
- revising sales procedures in terms of training and selection of distributors,
- wording and format of literature and the mechanics or any commission payments and claw back
- providing for the adequacy of staffing (numbers and competence)
- revising underwriting processes
- revising claims handling processes
- altering the capital allocation methodology
- improving the systems and data recording processes
- revising policy design
- improving wording of policy contracts.
- expense controls
- Policy retention activity

[4]

ii)

- a. The following could be the reasons for the increasing trend in the loss ratio:
- Renewing Bad Risk
 - Increase in general morbidity rates
 - Effect of initial underwriting
 - Medical treatment cost inflation/medical advances
 - Medical wage inflation
 - Waiting period and exclusions
 - Random fluctuations
 - Adverse profile mix
 - Outbreak of disease
 - Regulatory changes
 - Anti-selection effects/competitors providing cheaper premiums for better risks
 - Effects of changes in reinsurance (not sure if loss ratios are gross or net)

[4]

- b. The main limitation of using Policy Year as basis of monitoring the portfolio performance is that it takes longer time for experience to fully develop. Apart from this, policy year extends to two calendar years and, therefore, it is not very good measure to track frequency and severity trend, e. g., inflation.

[2]

[Total Marks-10]

Solution 9:

- i) The allowance of unexpired risk will consist of unexpired premium reserve (UPR) and unexpired risk reserves (URR).

Both will include the cost of expected future claims and expenses for unexpired portion.

First we need to set two assumptions:

1. How business is sold throughout the year
2. Length of cover provided by the policies

In the absence of any other information, it would be reasonable to assume that PMI business is sold evenly throughout the calendar year and as the premiums are annual, it is again reasonable to assume that cover is one year long.

An alternative, and more accurate approach, would be to use the actual unexpired period calculated on a policy-by-policy basis, in which case the last two assumptions would not be required.

Next, we need an assumption on the earnings of the premium, i.e. how the risk on the policy is spread over the year. One option would be to assume that it is even over the year. The UPR in this case would be a simple proportion of the written premium based on the number of days

cover unexpired (either assumed or actual, as above) as at the calculation date.

However, it is perhaps better argued that there will be more claims over the winter period (eg for pneumonia), in which case the risk is greater over those months. The UPR can be adjusted to allow for this by using a monthly approach to the calculation, putting more risk weight on the winter months.

There is also an increase of risk with age, as health deteriorates with age and claims are more likely at the end of the policy year. However, over the space of one year, this variation in risk is not great and could be taken to be negligible.

For an ongoing company, the UPR should be reduced for the uneven incidence of expenses over the year.

Also, we need to consider the purpose of the reserving exercise.

For example, if it is for statutory purposes, we would want a prudent method, whereas for internal purposes we would want a best estimate basis.

An approximate allowance for prudence could be built in, for example, by taking a larger premium for the calculation (even the full written premium, so that when we take the proportion to calculate the UPR, the result would be larger).

This will lower the reserve, the desirability of which depends again on the purpose of the calculation.

Finally, the reserve should be increased if necessary to allow for inadequate premiums, giving the additional unexpired risk reserve. In order to assess this, the adequacy of past premiums should be analyzed.

[7]

ii)

– **Improvements in recovery rates from cancer.**

Accelerated CI – If the said cancer is covered by the CI product, it will only impact the payment time. Hence, there is unlikely any change in the reserves. On other hand, if the said cancer is not covered in the policy terms and condition then it will not impact the reserves.

Income Protection – Improvement in recovery rates will reduce the expected payouts. Policyholders may be able to return to work early or for some part time basis. Hence, the reserves are expected to be reduce. Alternatively, benefits could be paid for a longer duration, for those cases where the claimant would otherwise have died; hence reserves could increase.

PMI - Some patients may incur lower treatment costs due to quicker full recovery, and others may incur higher treatment costs if they might otherwise have died.

– **Flood in the city**

Accelerated CI – IBNR reserve is expected to increase. However, the amount will not be material

Income Protection – catastrophe reserve will increase

PMI –The expected cost of claim will increase due to increase in hospital cost due to shortage of nurses/doctors/facilities. However, the cost rise will depend on whether there are fixed rate agreements in place between the insurer and hospitals. This will increase the reserves. The number of claims may also increase if the flood lead to increases in illness e.g. through breakdown in sanitation, influx of waterborne diseases.

– **Increase in the take-up rate of a diagnostic test for a serious cancer, following a high profile celebrity case.**

Accelerated CI – If the change caused diagnosis rates to increase, then accelerated critical illness pay outs could be brought forward, causing reserves to increase slightly.

Income Protection – If the treatment for the cancer was severe, this could increase claim rates. And if this is under policy term and condition then the reserves are likely to increase. On the other hand, if the cases are picked up early as a result of screening then treatment has started early before triggering to IP claim definition then the reserves could reduce.

PMI –This is likely to increase the claim rates (of cancer) and so PMI reserves will increase. Also, if the cost of screening is covered under PMI terms and condition, then reserves will also increase.

[9]

[Total Marks-16]

Solution 10:

i) Gross Sum Assured = 102mn

Net Sum Assured = $102 * (1-40\%) = 61.2\text{mn}$

Gross Reserves = 13mn

Net Reserves = 7.8mn

Gross Sum at risk = $102-13 = 89\text{mn}$

Net Sum at risk = $61.2-7.8 = 53.4\text{ mn}$

First Factor = 85%

Second Factor = 60%

$$\text{RSM} = 85\% * 13 + 60\% * 89 = 64.45\text{mn}$$

Items	INR Million
Policyholders' Assets	16
Policyholders' Liabilities	13
Policyholders' Surplus	3
Shareholders' Assets	201
Shareholders' Liabilities	2.5
Shareholders' Surplus	198.5
ASM	198.1
RSM	64.45
Solvency Ratio	307%

[3]

ii) Following are the four ways by which the company can improve its solvency position:

1. Take more reinsurance to take maximum credit in solvency capital: From the above solvency capital calculation, we came to know that company is not taking sufficient reinsurance to take maximum credit of reinsurance in solvency capital. The company may evaluate to increase quota share proportion from 40% to take maximum benefit from reinsurance. Further, the cost and terms of additional reinsurance should be calculated/evaluated before taking any decision.

2. Remove unnecessary MAD from reserves: The company can revisit the reserving methodology and assumptions to ensure that MAD is sufficient, not over.

3. Reduce Inadmissible Assets: Since inadmissible assets are not considered in ASM, hence minimize these type of assets will increase the solvency ratio.

4. Better investment strategy: Better investment strategy with the aim of higher investment return, lower default risk and matched assets with liabilities will help in improving solvency position for the company.

[4]

iii) Following steps should be taken:

Before we start we should decide on the projection period, say 5 years.

First of all, decide the projected amount of business in next few years, both from CI and IP business

The business will need to be split into appropriate model points that represent the business, split by average premium, sum assured, policy term, benefits etc.

The liabilities and assets would then be projected forward on assumptions that represent expected future experience.

For the assets, assume expected type of assets in which investments will be made. Project coupon payments and maturity value of these assets in the future projection period. Along with asset value, project future inadmissible assets too.

For the liabilities, project both CI and IP businesses for their expected payouts – benefits and expenses. These need to calculate on prudent basis.

The liabilities need to take into account future bonuses in IP product also in line with expected bonus setting philosophy.

The projected supervisory liabilities and assets can be valued at the end of each year. Then project available solvency margin for the projected period.

Project statutory required solvency margin for both projected new business and existing business, with expected reinsurance

Then for the projected period calculate solvency ratio as available solvency margin divided by required solvency margin.

Capital injection should be set such that solvency ratio at any time will meet minimum statutory requirement.

[5]

[Total Marks-12]
